PHYLLO / 144-point golden-angle audio display

Native review companion to the EasyEDA candidate — not a full EasyEDA project import.

5 circuit sheets / 167 components / explicit global-net connections

LED chain 001-036



文件: sheets/led-1.kicad_sch

LED chain 037-072



文件: sheets/led-2.kicad_sch

LED chain 073-108



文件: sheets/led-3.kicad_sch

LED chain 109-144



文件: sheets/led-4.kicad_sch

Power and interface



文件: sheets/power-interface.kicad_sch

STM32F411 development board and microphone are off-board modules. Their exact delivered identity is not frozen.

ERC does not validate reverse-polarity operation, current capacity, heat, pin polarity or wire harness assembly.

PCB routing is connected. External power intent is declared; final D121 DOUT is NC. Manufacturing review remains open.

PROTOTYPE DESIGN — NO HARDWARE TEST EVIDENCE

Sheet: /
File: phyllo-copper-review.kicad_sch

Title: PHYLLO / circuit review index

Size: A3	Date: 2026-10-03	**Rev: PROTO-02**
KiCad E.D.A. 10.0.5		Id: 1/6

PHYLLO / LED chain 001–036 / candidate pin-table schematic

Pin numbers/names from saved vendor library. Global labels join all sheets. No fabrication release.

D126
LED_DATA_IN
5V_LED
DIN
VDD
DOUT
VSS
DATA_001
GND
WS2812B-B-V6

D139
DATA_001
5V_LED
DIN
VDD
DOUT
VSS
DATA_002
GND
WS2812B-B-V6

D118
DATA_002
5V_LED
DIN
VDD
DOUT
VSS
DATA_003
GND
WS2812B-B-V6

D131
DATA_003
5V_LED
DIN
VDD
DOUT
VSS
DATA_004
GND
WS2812B-B-V6

D110
DATA_004
5V_LED
DIN
VDD
DOUT
VSS
DATA_005
GND
WS2812B-B-V6

D144
DATA_005
5V_LED
DIN
VDD
DOUT
VSS
DATA_006
GND
WS2812B-B-V6

D123
DATA_006
5V_LED
DIN
VDD
DOUT
VSS
DATA_007
GND
WS2812B-B-V6

D89
DATA_007
5V_LED
DIN
VDD
DOUT
VSS
DATA_008
GND
WS2812B-B-V6

D68
DATA_008
5V_LED
DIN
VDD
DOUT
VSS
DATA_009
GND
WS2812B-B-V6

D102
DATA_009
5V_LED
DIN
VDD
DOUT
VSS
DATA_010
GND
WS2812B-B-V6

D136
DATA_010
5V_LED
DIN
VDD
DOUT
VSS
DATA_011
GND
WS2812B-B-V6

D115
DATA_011
5V_LED
DIN
VDD
DOUT
VSS
DATA_012
GND
WS2812B-B-V6

D94
DATA_012
5V_LED
DIN
VDD
DOUT
VSS
DATA_013
GND
WS2812B-B-V6

D128
DATA_013
5V_LED
DIN
VDD
DOUT
VSS
DATA_014
GND
WS2812B-B-V6

D141
DATA_014
5V_LED
DIN
VDD
DOUT
VSS
DATA_015
GND
WS2812B-B-V6

D120
DATA_015
5V_LED
DIN
VDD
DOUT
VSS
DATA_016
GND
WS2812B-B-V6

D107
DATA_016
5V_LED
DIN
VDD
DOUT
VSS
DATA_017
GND
WS2812B-B-V6

D86
DATA_017
5V_LED
DIN
VDD
DOUT
VSS
DATA_018
GND
WS2812B-B-V6

D65
DATA_018
5V_LED
DIN
VDD
DOUT
VSS
DATA_019
GND
WS2812B-B-V6

D44
DATA_019
5V_LED
DIN
VDD
DOUT
VSS
DATA_020
GND
WS2812B-B-V6

D31
DATA_020
5V_LED
DIN
VDD
DOUT
VSS
DATA_021
GND
WS2812B-B-V6

D52
DATA_021
5V_LED
DIN
VDD
DOUT
VSS
DATA_022
GND
WS2812B-B-V6

D73
DATA_022
5V_LED
DIN
VDD
DOUT
VSS
DATA_023
GND
WS2812B-B-V6

D81
DATA_023
5V_LED
DIN
VDD
DOUT
VSS
DATA_024
GND
WS2812B-B-V6

D60
DATA_024
5V_LED
DIN
VDD
DOUT
VSS
DATA_025
GND
WS2812B-B-V6

D39
DATA_025
5V_LED
DIN
VDD
DOUT
VSS
DATA_026
GND
WS2812B-B-V6

D26
DATA_026
5V_LED
DIN
VDD
DOUT
VSS
DATA_027
GND
WS2812B-B-V6

D18
DATA_027
5V_LED
DIN
VDD
DOUT
VSS
DATA_028
GND
WS2812B-B-V6

D10
DATA_028
5V_LED
DIN
VDD
DOUT
VSS
DATA_029
GND
WS2812B-B-V6

D23
DATA_029
5V_LED
DIN
VDD
DOUT
VSS
DATA_030
GND
WS2812B-B-V6

D36
DATA_030
5V_LED
DIN
VDD
DOUT
VSS
DATA_031
GND
WS2812B-B-V6

D57
DATA_031
5V_LED
DIN
VDD
DOUT
VSS
DATA_032
GND
WS2812B-B-V6

D78
DATA_032
5V_LED
DIN
VDD
DOUT
VSS
DATA_033
GND
WS2812B-B-V6

D99
DATA_033
5V_LED
DIN
VDD
DOUT
VSS
DATA_034
GND
WS2812B-B-V6

D133
DATA_034
5V_LED
DIN
VDD
DOUT
VSS
DATA_035
GND
WS2812B-B-V6

D112
DATA_035
5V_LED
DIN
VDD
DOUT
VSS
DATA_036
GND
WS2812B-B-V6

PROTOTYPE DESIGN — NO HARDWARE TEST EVIDENCE

Sheet: /LED chain 001–036/
File: led-1.kicad_sch

Title: LED chain 001–036

Size: A3	Date: 2026-10-03	**Rev: PROTO-02**
KiCad E.D.A. 10.0.5		Id: 2/6

PHYLLO / LED chain 037–072 / candidate pin-table schematic

Pin numbers/names from saved vendor library. Global labels join all sheets. No fabrication release.

D91
DATA_036 4 DIN DOUT 2 DATA_037
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D70
DATA_037 4 DIN DOUT 2 DATA_038
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D49
DATA_038 4 DIN DOUT 2 DATA_039
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D62
DATA_039 4 DIN DOUT 2 DATA_040
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D83
DATA_040 4 DIN DOUT 2 DATA_041
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D104
DATA_041 4 DIN DOUT 2 DATA_042
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D125
DATA_042 4 DIN DOUT 2 DATA_043
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D138
DATA_043 4 DIN DOUT 2 DATA_044
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D117
DATA_044 4 DIN DOUT 2 DATA_045
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D130
DATA_045 4 DIN DOUT 2 DATA_046
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D143
DATA_046 4 DIN DOUT 2 DATA_047
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D122
DATA_047 4 DIN DOUT 2 DATA_048
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D135
DATA_048 4 DIN DOUT 2 DATA_049
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D114
DATA_049 4 DIN DOUT 2 DATA_050
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D93
DATA_050 4 DIN DOUT 2 DATA_051
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D127
DATA_051 4 DIN DOUT 2 DATA_052
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D140
DATA_052 4 DIN DOUT 2 DATA_053
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D119
DATA_053 4 DIN DOUT 2 DATA_054
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D132
DATA_054 4 DIN DOUT 2 DATA_055
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D98
DATA_055 4 DIN DOUT 2 DATA_056
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D77
DATA_056 4 DIN DOUT 2 DATA_057
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D56
DATA_057 4 DIN DOUT 2 DATA_058
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D111
DATA_058 4 DIN DOUT 2 DATA_059
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D124
DATA_059 4 DIN DOUT 2 DATA_060
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D137
DATA_060 4 DIN DOUT 2 DATA_061
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D116
DATA_061 4 DIN DOUT 2 DATA_062
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D95
DATA_062 4 DIN DOUT 2 DATA_063
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D74
DATA_063 4 DIN DOUT 2 DATA_064
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D53
DATA_064 4 DIN DOUT 2 DATA_065
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D40
DATA_065 4 DIN DOUT 2 DATA_066
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D27
DATA_066 4 DIN DOUT 2 DATA_067
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D61
DATA_067 4 DIN DOUT 2 DATA_068
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D82
DATA_068 4 DIN DOUT 2 DATA_069
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D103
DATA_069 4 DIN DOUT 2 DATA_070
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D90
DATA_070 4 DIN DOUT 2 DATA_071
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

D69
DATA_071 4 DIN DOUT 2 DATA_072
5V_LED 1 VDD VSS 3 GND
WS2812B-B-V6

PROTOTYPE DESIGN — NO HARDWARE TEST EVIDENCE

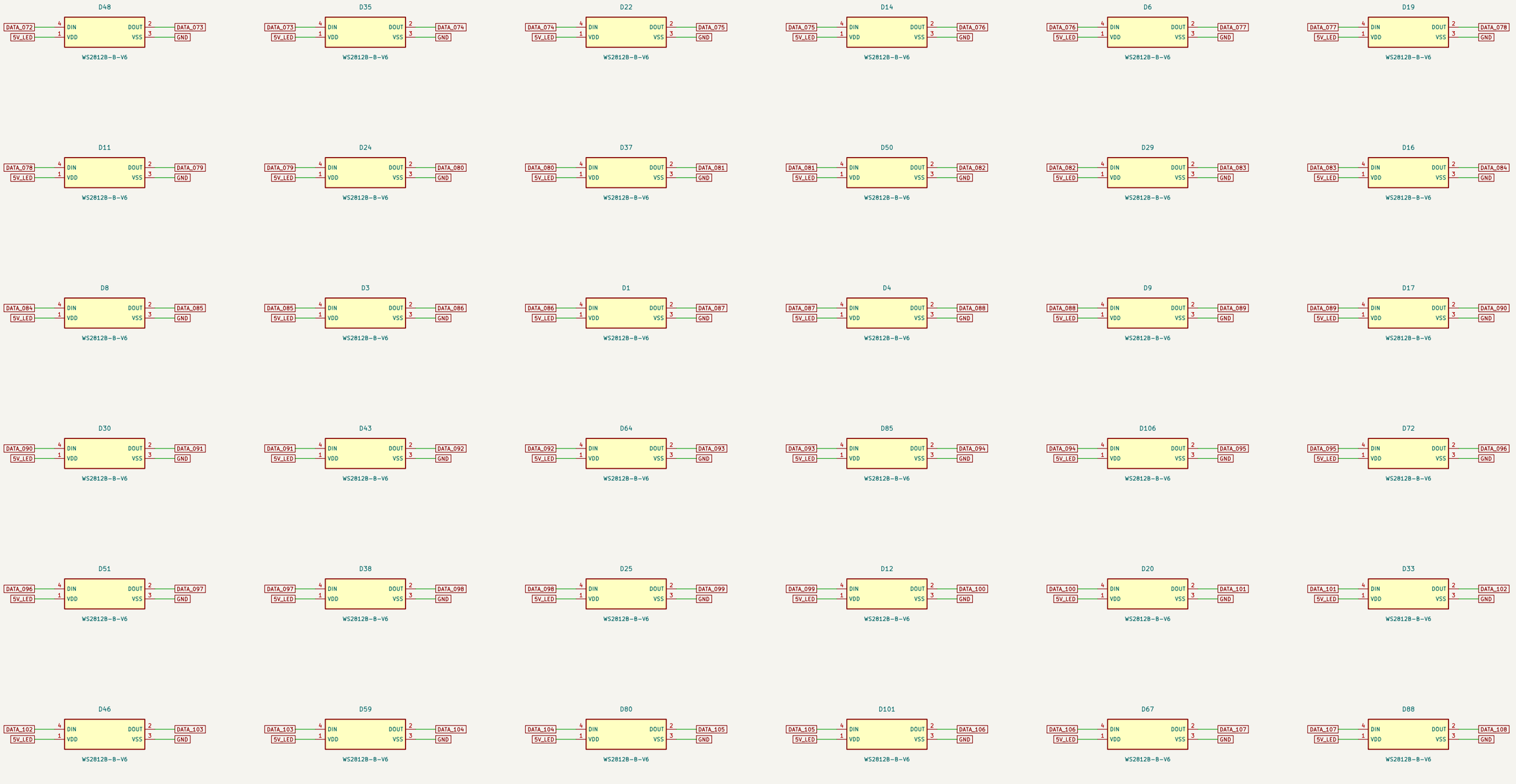
Sheet: /LED chain 037–072/
File: led-2.kicad_sch

Title: LED chain 037–072

Size: A3	Date: 2026-10-03	**Rev: PROTO-02**
KiCad E.D.A. 10.0.5		Id: 3/6

PHYLLO / LED chain 073–108 / candidate pin-table schematic

Pin numbers/names from saved vendor library. Global labels join all sheets. No fabrication release.



PROTOTYPE DESIGN – NO HARDWARE TEST EVIDENCE

Sheet: /LED chain 073–108/
File: led–3.kicad_sch

Title: LED chain 073–108

Size: A3	Date: 2026–10–03	**Rev: PROTO–02**
KiCad E.D.A. 10.0.5		Id: 4/6

PHYLLO / LED chain 109–144 / candidate pin-table schematic

Pin numbers/names from saved vendor library. Global labels join all sheets. No fabrication release.



PROTOTYPE DESIGN – NO HARDWARE TEST EVIDENCE

Sheet: /LED chain 109–144/
File: led-4.kicad_sch

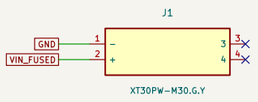
Title: LED chain 109–144

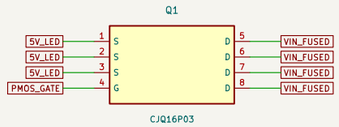
Size: A3	Date: 2026-10-03	**Rev: PROTO-02**
KiCad E.D.A. 10.0.5		Id: 5/6

PHYLLO / power and controller interface / candidate pin tables

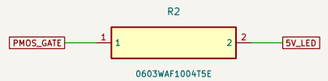
External fused 5V only. PMOS D=input, S=protected rail. NEVER parallel controller USB VBUS with external 5V.

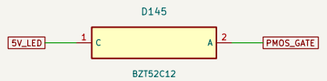
C1–C4/C6–C9: Chemi-Con PXF 270uF/10V, pin1=positive. Physical harness and power tests remain required. ERC is not circuit simulation.



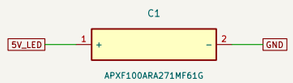


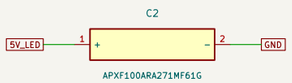


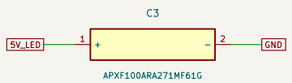


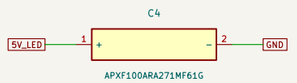


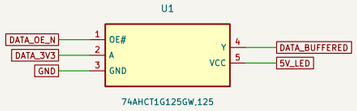


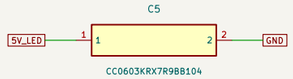


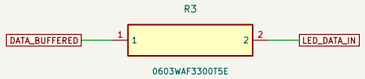






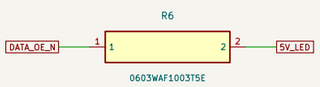


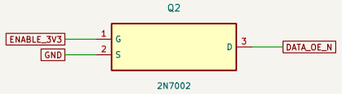




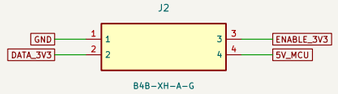




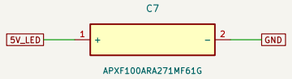


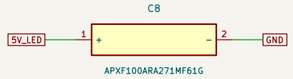


















PWR_FLAG declares intended external supply through J1/Q1, not hardware proof. Last LED D121 DOUT is intentionally NC.

PROTOTYPE DESIGN — NO HARDWARE TEST EVIDENCE

Sheet: /Power and interface/
File: power-interface.kicad_sch

Title: Power and controller interface

Size: A3	Date: 2026-10-03	**Rev: PROTO-02**
KiCad E.D.A. 10.0.5		Id: 6/6